



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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EXPERIMENT :5

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1. Aim

i) Performance Benchmarking : Normal View vs. Materialized View (Medium)

1. Create a large dataset:

- Create a table names transaction_data (id, value) with 1 million records.
- take id 1 and 2, and for each id, generate 1 million records in value column
- Use Generate series () and random() to populate the data.

2. Create a normal view and materialized view to for sales summary, which includes total quantity_sold, total sales, and total orders with aggregation.

3. Compare the performance and execution time of both

ii) Views: Securing Data Access with Views and Role-Based Permissions (Hard)

The company TechMart Solutions stores all sales transactions in a central database. A new reporting team has been formed to analyze sales but they should not have direct access to the base tables for security reasons. The database administrator has decided to:

1. Create restricted views to display only summarized, non-sensitive data.
2. Assign access to these views to specific users using DCL commands (GRANT, REVOKE).

2. Tools Used: SQL Server Management Studio

3. Query

Q1)

```
CREATE TABLE transaction_data (  
    id INT,  
    value INT  
);
```



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```
INSERT INTO transaction_data (id, value)
```

```
SELECT 1, random() * 1000
```

```
FROM generate_series(1, 1000000);
```

```
INSERT INTO transaction_data (id, value)
```

```
SELECT 2, random() * 1000
```

```
FROM generate_series(1, 1000000);
```

```
SELECT *FROM transaction_data;
```

```
CREATE OR REPLACE VIEW sales_summary_view AS
```

```
SELECT
```

```
    id,
```

```
    COUNT(*) AS total_orders,
```

```
    SUM(value) AS total_sales,
```

```
    AVG(value) AS avg_transaction
```

```
FROM transaction_data
```

```
GROUP BY id;
```

```
CREATE MATERIALIZED VIEW sales_summary_mv AS
```

```
SELECT
```

```
    id,
```

```
    COUNT(*) AS total_orders,
```

```
    SUM(value) AS total_sales,
```

```
    AVG(value) AS avg_transaction
```

```
FROM transaction_data
```

```
GROUP BY id;
```

```
REFRESH MATERIALIZED VIEW sales_summary_mv;
```

```
EXPLAIN ANALYZE SELECT * FROM sales_summary_view;
```

```
EXPLAIN ANALYZE SELECT * FROM sales_summary_mv;
```

Q2)

```
CREATE TABLE customer_master (  
    customer_id VARCHAR(5) PRIMARY KEY,  
    full_name VARCHAR(50) NOT NULL,  
    phone VARCHAR(15),  
    email VARCHAR(50),  
    city VARCHAR(30)  
);  
  
CREATE TABLE product_catalog (  
    product_id VARCHAR(5) PRIMARY KEY,  
    product_name VARCHAR (50) NOT NULL,  
    brand VARCHAR(30),  
    unit_price NUMERIC(10,2) NOT NULL  
);  
  
CREATE TABLE sales_orders (  
    order_id SERIAL PRIMARY KEY,  
    product_id VARCHAR(5) REFERENCES product_catalog(product_id),  
    quantity INT NOT NULL,  
    customer_id VARCHAR(5) REFERENCES customer_master(customer_id),  
    discount_percent NUMERIC(5,2),  
    order_date DATE NOT NULL  
);  
  
INSERT INTO customer_master (customer_id, full_name, phone, email, city) VALUES  
( 'C1', 'Amit Sharma', '9876543210', 'amit.sharma@example.com', 'Delhi'),  
( 'C2', 'Priya Verma', '9876501234', 'priya.verma@example.com', 'Mumbai' ),  
( 'C3', 'Ravi Kumar', '9988776655', 'ravi.kumar@example.com', 'Bangalore' ),  
( 'C4', 'Neha Singh', '9123456789', 'neha.singh@example.com', 'Kolkata'),  
( 'C5', 'Arjun Mehta', '9812345678', 'arjun.mehta@example.com', 'Hyderabad' ),  
( 'C6', 'Sneha Reddy', '9090909090', 'sneha.reddy@example.com', 'Chennai' ),  
( 'C7', 'Vikram Das', '9123412345', 'vikram.das@example.com', 'Pune' ),
```

```
('C8', 'Rohit Gupta', '9000000001' , 'rohit. gupta@example.com', 'Lucknow' ) ,  
( 'C9', 'Pooja Nair', '9898989898', 'pooja. nair@example.com', 'Kochi'),  
( 'C10', 'Ankit Yadav', '9345678901', 'ankit.yadav@example.com' , 'Ahmedabad' ) ;
```

```
INSERT INTO product_catalog (product_id, product_name, brand, unit_price) VALUES
```

```
('P1', 'Smartphone X100', 'Samsung', 25000.00),
```

```
('P2', 'Laptop Pro 15', 'Dell', 65000.00),
```

```
('P3', 'Wireless Earbuds', 'Sony', 5000.00),
```

```
('P4', 'Smartwatch Fit', 'Apple', 30000.00),
```

```
('P5', 'Tablet 10.5', 'Lenovo', 22000.00),
```

```
('P6', 'Gaming Console', 'Sony', 45000.00),
```

```
('P7', 'Bluetooth Speaker', 'JBL', 7000.00),
```

```
('P8', 'Digital Camera', 'Canon', 55000.00),
```

```
('P9', 'LED TV 55 inch', 'LG', 60000.00),
```

```
('P10', 'Power Bank 20000mAh', 'Mi', 2500.00);
```

```
INSERT INTO sales_orders (product_id, quantity, customer_id, discount_percent, order_date)  
VALUES
```

```
('P1', 2, 'C1', 5.00, '2025-09-01'),
```

```
('P2', 1, 'C2', 10.00, '2025-09-02'),
```

```
('P3', 3, 'C3', 0.00, '2025-09-03'),
```

```
('P4', 1, 'C4', 8.00, '2025-09-04'),
```

```
('P5', 2, 'C5', 5.00, '2025-09-05'),
```

```
('P6', 1, 'C1', 12.00, '2025-09-06'),
```

```
('P7', 2, 'C2', 0.00, '2025-09-07'),
```

```
('P8', 1, 'C3', 10.00, '2025-09-08'),
```

```
('P9', 1, 'C6', 15.00, '2025-09-09'),
```

```
('P10', 4, 'C7', 0.00, '2025-09-10'),
```

```
('P1', 1, 'C8', 5.00, '2025-09-11'),
```

```
('P2', 2, 'C9', 10.00, '2025-09-12') ,
```

```
('P3', 2, 'C10', 0.00, '2025-09-13'),
```

```
('P4', 1, 'C5', 8.00, '2025-09-14'),
```

```
('P5', 3, 'C6', 5.00, '2025-09-15'),  
( 'P6', 1, 'C7', 12.00, '2025-09-16'),  
( 'P7', 2, 'C8', 0.00, '2025-09-17'),  
( 'P8', 1, 'C9', 10.00, '2025-09-18'),  
( 'P9', 1, 'C10', 15.00, '2025-09-19'),  
( 'P10', 5, 'C4', 0.00, '2025-09-20');  
  
CREATE VIEW vW_ORDER_SUMMARY AS  
  
SELECT  
  
    O.order_id,  
  
    O.order_date,  
  
    P.product_name,  
  
    C.full_name,  
  
    (P.unit_price * O.quantity)  
    - ((P.unit_price * O.quantity) * O.discount_percent / 100) AS final_cost  
  
FROM customer_master AS C  
  
JOIN sales_orders AS O  
  
    ON O.customer_id = C.customer_id  
  
JOIN product_catalog AS P  
  
    ON P.product_id = O.product_id;  
  
  
SELECT * FROM vW_ORDER_SUMMARY;  
  
CREATE ROLE piyush LOGIN PASSWORD 'piyush123';  
  
GRANT SELECT ON vW_ORDER_SUMMARY TO piyush;  
  
REVOKE SELECT ON vW_ORDER_SUMMARY FROM piyush;
```

4. Outputs

Showing rows: 1 to 1000 Page No: 1 of 2000

	id integer	value integer
1	1	305
2	1	617
3	1	797
4	1	790
5	1	83
6	1	631

✓ Successfully run. Total query runtime: 655 msec. 2000000 rows affected.

Data Output Messages Notifications

Showing rows: 1 to 40 Page No: 1 of 1

	order_id integer	order_date date	product_name character varying (50)	full_name character varying (50)	final_cost numeric
1	1	2025-09-01	Smartphone X100	Amit Sharma	47500.000000000000000000
2	2	2025-09-02	Laptop Pro 15	Priya Verma	58500.000000000000000000
3	3	2025-09-03	Wireless Earbuds	Ravi Kumar	15000.000000000000000000
4	4	2025-09-04	Smartwatch Fit	Neha Singh	27600.000000000000000000
5	5	2025-09-05	Tablet 10.5	Arjun Mehta	41800.000000000000000000
6	6	2025-09-06	Gaming Console	Amit Sharma	39600.000000000000000000
7	7	2025-09-07	Bluetooth Speaker	Priya Verma	14000.000000000000000000
8	8	2025-09-08	Digital Camera	Ravi Kumar	49500.000000000000000000
9	9	2025-09-09	LED TV 55 inch	Sneha Reddy	51000.000000000000000000

Total rows: 40 Query complete 00:00:00.087 LF Ln 90, Col 32

5. Learning Outcomes

- Learned how to create database views to present summarized and non-sensitive data while hiding underlying table details.
- Understood the difference between Normal Views (executed dynamically) and Materialized Views (pre-computed and stored).
- Gained practical knowledge of how to create users/roles and grant or revoke permissions on views to ensure controlled access in multi-user database environments.